MODULE 03

Data Visualization

DATA SCIENCE · LEARNING NOTES

Mohan Babu University · Course-aligned study guide

Concepts • examples • practice • revision

01. Pandas plotting

Learning focus. This section introduces pandas plotting as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to pandas plotting. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

02. Seaborn introduction

Learning focus. This section introduces seaborn introduction as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to seaborn introduction. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

03. Line plots

Learning focus. This section introduces line plots as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to line plots. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

04. Bar plots

Learning focus. This section introduces bar plots as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to bar plots. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

05. Histograms

Learning focus. This section introduces histograms as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to histograms. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

06. Density plots

Learning focus. This section introduces density plots as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to density plots. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

07. Scatter plots

Learning focus. This section introduces scatter plots as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to scatter plots. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

08. Facet grids

Learning focus. This section introduces facet grids as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to facet grids. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

09. Categorical data

Learning focus. This section introduces categorical data as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to categorical data. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

10. Chart selection

Learning focus. This section introduces chart selection as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to chart selection. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

11. Labels and annotations

Learning focus. This section introduces labels and annotations as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to labels and annotations. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

12. Avoiding misleading charts

Learning focus. This section introduces avoiding misleading charts as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to avoiding misleading charts. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

13. Visualization exercise

Learning focus. This section introduces visualization exercise as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to visualization exercise. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?

14. Module review

Learning focus. This section introduces module review as part of data visualization. Connect the idea to a concrete dataset, identify the question being answered, and explain what evidence would support a conclusion.

Concepts and workflow

Start by defining the problem and inspecting the available data. Check data types, missing values, unusual observations, and assumptions before applying a method. Select an approach that matches the question and document each transformation so the analysis can be reproduced.

Practical example

Imagine a small dataset with a few rows and columns relevant to module review. Inspect the values, summarize the important patterns, and compare the result before and after applying the technique. Explain both the result and its limitations in plain language.

Practice checklist

- State the goal and expected output.
- Choose suitable data and tools.
- Run the method and inspect the result.
- Record assumptions, limitations, and next steps.

Review question: When is this technique appropriate, and what could make its result misleading?